

Robust damping improvement against the vortex-induced vibration in flexible bridges using multiple tuned mass damper inerters

Jun Dai^{a,b}, Cen Yang^{a,b}, Pan-Pan Gai^c, Zhao-Dong Xu^{a,b,*}, Xiao Yan^d, Wen-Peng Xu^{a,b}

^a China-Pakistan Belt and Road Joint Laboratory on Smart Disaster Prevention of Major Infrastructures, Southeast University, Nanjing 210096, China

^b School of Civil Engineering, Southeast University, Nanjing 210096, China

^c Faculty of Civil Engineering and Mechanics, Jiangsu University, Zhenjiang 212013, China

^d School of Urban Planning and Municipal Engineering, Xi'an Polytechnic University, Xi'an 710048, China

ARTICLE INFO

Keywords:

Vortex-induced vibration
Multiple tuned mass damper inerter
Flexible bridge
Robustness

ABSTRACT

The tuned mass damper inerter (TMDI) is considered as one of countermeasures to mitigate the very-low-frequency vertical bending vortex-induced vibration (VIV) in flexible bridges. However, the control robustness of the TMDI against the system uncertainty is a serious concern due to the limited inerter span capacity. In this paper, the multiple tuned mass damper inerter (MTMDI) is used instead of identical TMDIs to provide a robust equivalent damping ratio against the uncertain VIV within the entire lock-in wind speed range. A design framework of the MTMDI considering the inerter span length is proposed based on a universal VIV model suitable for the fully closed, centrally slotted, and semi-closed box decks. In the framework, the fluctuation of VIV characteristics with the wind speed, the probability of occurrence of the VIV, and the uncertainties in VIV characteristics and deck dynamics are considered, simultaneously. Two objection functions are defined to focus on the maximizing the equivalent damper ratio and reducing the VIV mitigation failure, respectively. Finally, numerical analysis of a flexible bridge-MTMDI system subjected to the VIV is performed. The results show that the wind speed probability distribution plays a role of weighting coefficient in the MTMDI optimization. The choice of the objective function depends on the mass ratio of the MTMDI and the level of the system uncertainty. The well-optimized MTMDI achieved a more robust control result than the TMDI within the entire lock-in wind speed range.

1. Introduction

The vortex-induced vibration (VIV) is a well-known vibration in engineering structures such as flexible bridges, high buildings, towers, pipes, cables, and chimneys [1,2]. The VIVs in bridges may occur on the deck, hanger, tower, and cable [3]. Among them, the VIVs in decks seems to be paid more attention due to remarkable cross-section configurations and load-bearing role of the deck. Moreover, the VIVs in decks were widely observed in real bridges around the world. Table 1 summarizes the VIVs observed in real bridges [4–17]. These flexible bridges had low natural frequencies and damping ratios, accordingly the large-amplitude VIVs occurred at modest wind speeds or even low wind speeds.

The VIV in decks is considered as a self-limited vibration, thus it has not led to a catastrophic event so far, but it will result in the driving

discomfort, people panic, and fatigue problems of structural components [18]. Therefore, some measures are taken to mitigate the VIV in decks, they are aerodynamic countermeasures and tuned-based dampers [19]. Fujino and Siringoringo summarized that the understanding vibration mechanisms is the most important step in finding a suitable cure for the vibration problem of long-span bridges [20]. On the basis of understanding the VIV mechanism, various aerodynamic countermeasures were designed to avoid the periodic vortex shedding, such as the fairing, horizontal plate, baffle, guide vanes, flaps, spoiler, and deflector [19–21]. However, the aerodynamic countermeasure may not be effective in mitigating the VIV if the cross section of the deck changes a bit, in other words, the aerodynamic countermeasure is not universally effective for the VIV mitigation [22]. Alternatively, the tuned-based dampers such as the tuned mass damper (TMD) [23], multiple tuned mass dampers (MTMD) [24], nonlinear energy sink [25], semi-active

* Corresponding author at: China-Pakistan Belt and Road Joint Laboratory on Smart Disaster Prevention of Major Infrastructures, Southeast University, Nanjing 210096, China.

E-mail address: zhdxu@163.com (Z.-D. Xu).

<https://doi.org/10.1016/j.engstruct.2024.118221>

Received 20 October 2023; Received in revised form 25 April 2024; Accepted 16 May 2024

Available online 28 May 2024

0141-0296/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1

The VIVs observed in real bridges.

Type	Bridge name	Main span length (m)	Wind speed interval (m/s)	Vortex-induced amplitude (mm)
Cable-stayed bridge	Kessock Bridge[4]	240	23–25	110
	Second Severn Bridge[5]	456	15–20	192
	Ishikari Kako Bridge[6]	160	10	100
	Jindo Bridge[7]	340	9.8–11.5	100
	Wyc Bridge[8]	235	7–8	36
	Fred Hartman Bridge[9]	380	8–12	-
	Xihoumen Bridge[10]	1650	5–12	240
Suspension bridge	Humen Bridge[3]	888	8–10	190
	Great Belt East Bridge[11]	1624	5–12	320
	Tacoma Narrows Bridge[12]	853	< 18.4	-
	Deer Isle Bridge[13]	329	9.11	55
	Rio-Niteroi Bridge[14]	300	15–17	250
Beam bridge	Trans-Tokyo Bay Bridge[15]	240	16–17	540
	Wolga Bridge[16]	155	11.6–15.6	560–700
	Great Belt East Bridge approach spans[17]	193	16–20	100

TMD [26,27], and active TMD [28] were used to mitigate the vertical-bending VIV in decks numerically, experimentally, and realistically. Among these tuned-based dampers, the TMD and MTMD are considered the most mature technologies owing to the clear working principle and easy configuration. The Trans-Tokyo Bay Bridge [15], Rio-Niteroi Bridge [14], Kessock Bridge [29], Volgograd Bridge [30], Jindo Bridge [7], Chongqi Bridge [31], and Hong Kong-Zhuhai-Macao Bridge deep-water non-navigable bridge [23] all adopted the TMD or MTMD scheme to improve the equivalent damping ratio of the deck against the VIV.

The large static stretching and dynamic stroke of the auxiliary mass is a significant but easily overlooked problem in the vertical bending VIV mitigation of flexible bridges [32]. The problem is well understood that the flexible bridges often have very low natural frequencies or vibration frequencies, accordingly the TMD has a very low natural frequency [33].

According to $\Delta = g/(4\pi^2 f^2)$, for example a TMD with a nature frequency f of 0.5 Hz has a static stretching Δ of 1 m. As a result, it is very difficult to install the very-low-frequency TMD in the long-span bridge with a streamlined flat box deck. Furthermore, such streamlined flat box decks are more likely to suffer the large-amplitude VIV at low wind speeds [34]. The inerter devices with characteristics of high-static and low-dynamic stiffness provide a possible solution for the very-low-frequency TMD. Inspired by using the tuned mass damper inerter (TMDI) to control the wind-excited tall building [35], Dai et al. used the TMDI to mitigate the vertical bending VIV in flexible bridges, and they numerically found that the negative stiffness and mass amplification effects contributed by the inerter improve the control effect and also reduce the static stretching and dynamic stroke [36]. Xu et al. also used the TMDI to mitigate the undesired VIV in long-span bridges and proposed an optimal design method for the TMDI control [37]. Yu et al. used the nonlinear energy sink inerter to mitigate the VIV in bridges, and they found that the added inerter in general improved the control effect [38]. In flexible bridges, a completely inerter-ungrounded TMDI with easy implementability has worse effectiveness than a TMD, while an ideal inerter-grounded TMDI with poor implementability has much better effectiveness than a TMD [35–38]. In other words, the inerter span length plays a significant role in balancing the effectiveness and implementability of the MTMDI. Dai et al., Xu et al., and Yu et al. proposed analytical or numerical solutions of the optimal TMDI parameters considering the inerter span length in the flexible bridges [39–41,38]. Besides, the aerodynamic parameters fluctuating with the wind speed easily cause the variable VIV characteristics such as the amplitude and frequency [42,43]. As a result, the frequency-sensitive TMDI will partially lose its effectiveness outside the targeted wind speed, i.e., the vortex-induced resonance wind speed, especially for the TMDI with a small inerter span length. This is because the TMDI with identical design parameters focuses on a vortex-induced

resonance mitigation rather than a generalized VIV mitigation within the entire lock-in wind speed range. Furthermore, there are significant uncertainties in VIV characteristics and deck dynamics due to the complex fluid-structure interaction, modeling errors, environmental disturbances [44–47], accordingly the sensitivity concern of the TMDI become more serious. A simple way to solve this problem is to use the multiple tuned mass damper inerter (MTMDI) instead of the TMDI to mitigate the vertical bending VIV in decks. Such multiple-based strategies are widely adopted in the TMD control to improve the robustness against various frequency-related fluctuations and the effectiveness in the broadband vibration reduction or multi-mode vibration suppression [48,49]. Recently, the MTMDI has been used to reduce the seismic responses of adjacent buildings [50] and the multi-mode VIV in long-span bridges [51].

In this paper, the MTMDI is used to provide a robust equivalent damping ratio for the flexible bridge against the uncertain vertical bending VIV within the entire lock-in wind speed range, at the same time, the effect of the inerter span length on the MTMDI control is considered. Section 2 establishes dimensionless equations of motion of the deck-MTMDI system based on a universal nonlinear mathematical model of the vortex-induced force. Section 3 provides an approximate analytical solution of responses of the deck-MTMDI system using the Krylov-Bogoliubov method. Section 4 proposes a design framework of the MTMDI based on the defined equivalent damping ratio, in which the fluctuation of VIV characteristics with the wind speed, the probability of occurrence of the VIV, and the uncertainties in VIV characteristics and deck dynamics are considered, simultaneously. Section 5 presents a numerical example to demonstrate the superiority of the MTMDI control.

2. The mathematical modeling of deck-MTMDI system

In order to mitigate the very-low-frequency vertical VIV in the flexible bridge, the MTMDI is installed in the deck, as shown in Fig. 1. The equations of motion of the deck-MTMDI system subjected to the VIV at any section x are given by

$$m(x)\frac{\partial^2 y(x,t)}{\partial t^2} + c(x)\frac{\partial y(x,t)}{\partial t} + \frac{\partial^2 y(x,t)}{\partial x^2} \left[EI(x)\frac{\partial^2 y(x,t)}{\partial x^2} \right] = f_V(\dot{x}, x, t) + f_T(x, t) \quad (1)$$

$$m_n[\ddot{y}_n(t) + \ddot{y}(x_n, t)] + b_n[\ddot{y}_n(t) + \ddot{y}(x_n, t) - \ddot{y}(x_{nb}, t)] + c_n\dot{y}_n(t) + k_n y_n(t) = 0 \quad (2)$$

where y is the displacement of the deck; c and EI are the damping coefficient and rigidity of the deck; f_V is the vortex-induced force acting on the deck; f_T is the force provided by the MTMDI; y_n is the displacement

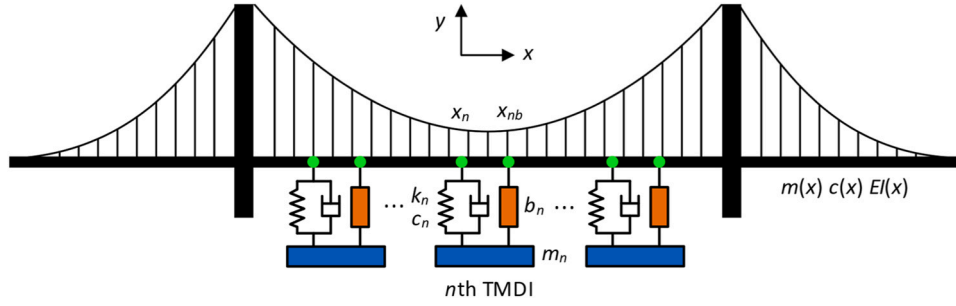


Fig. 1. The flexible bridge equipped with the MTMDI.

of the n th TMDI relative to deck displacement at the section x_n ; x_{nb} is the installation section position of the n th inerter; m_n , b_n , c_n , and k_n are the mass, inertance, damping, and stiffness of the n th TMDI. The force provided by the n th TMDI at different sections has different forms

$$f_T(x_n, t) = c_n \dot{y}_n(t) + k_n y_n(t) \quad (3)$$

$$f_T(x_{nb}, t) = b_n [\ddot{y}_n(t) + \ddot{y}(x_n, t) - \ddot{y}(x_{nb}, t)] \quad (4)$$

In the VIV of the rectangular cylinder, the mathematical models based on displacement-related nonlinear aerodynamic damping, such as the Scanlan model, van der Pol model, and Larsen model, perform well in describing the vortex-induced force [52]. However, in the vertical VIV of the bridge deck, the deck velocity should be included in the auto variables of the nonlinear aerodynamic damping rather than the deck displacement [42,43]. Therefore, Zhu et al. proposed three nonlinear mathematical models with the auto variable of deck velocity to describe the VIV in fully closed, centrally slotted, and semi-closed box decks, respectively [43]. If we focus on the stable amplitude of the deck in the VIV, the three nonlinear mathematical models are reduced as a universal VIV model [43], which is expressed as

$$f_V(\dot{x}, x, t) = \frac{1}{2} \rho U^2 (2D) \left[Y_1 \left(1 - \varepsilon_N \frac{\partial^2 y(x, t)}{\partial t^2 U^2} \right) \frac{\partial y(x, t)}{\partial t U} + Y_2 \frac{y(x, t)}{D} \right] \quad (5)$$

where ρ is the air density; U is the wind speed; D is the characteristic dimension of the deck; $K = \omega D/U$ is the reduced frequency; ω is the oscillation circular frequency. Y_1 , ε_N , and Y_2 are the aerodynamic parameters, which are function of K , thus Eq. (5) can describe the VIV within the entire lock-in wind speed range.

Introducing $y(x, t) = \Phi(x)q(t)$, and integrating from 0 to the deck length L at each side of Eq. (1), we obtain a modal equation

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{F}_V + \mathbf{F}_T \quad (6)$$

where $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the modal mass, damping, and stiffness matrices of the deck; $\mathbf{F}_V$ and $\mathbf{F}_T$ are the vortex-induced force matrix and the MTMDI force matrix, respectively. Here the mode coupling or the nonclassical damping induced by the added MTMDI is neglected, thus the deck-MTMDI system is simplified as a two-degree-of-freedom system. Then, Eq. (6) is reduced as

$$M\ddot{q} + C\dot{q} + Kq = F_V + F_T \quad (7)$$

Similarly, the modal equation of the n th TMDI is obtained by introducing $y_n(t) = \phi(x_n)q_n$

$$m_n(\ddot{q}_n + \ddot{q}) + b_n(\ddot{q}_n + \ddot{q} - \kappa_n \ddot{q}) + c_n \dot{q}_n + k_n q_n = 0 \quad (8)$$

After adding the inerter into the TMDI, the static stretching reduction of the n th auxiliary mass is $1/(1+\lambda_n)$, in which $\lambda_n = b_n/m_n$ and $\kappa_n = \phi_{nb}/\phi_n$. Furthermore, introducing $\omega_n = \sqrt{k_n/m_n}$ and $\xi_n = c_n/2m_n\omega_n$, F_V and F_T are rewritten as

$$F_V = \int_0^L \phi(x) f_V(\dot{x}, x, t) dx = \rho U^2 D \left(\alpha_2 Y_1 \frac{\dot{q}}{U} - \alpha_4 Y_1 \varepsilon_N \frac{\dot{q}^3}{U^3} + \alpha_2 Y_2 \frac{q}{D} \right) \quad (9)$$

$$F_T = \sum_{n=1}^N \phi_n f_T(x_n, t) = \sum_{n=1}^N \phi_n^2 m_n [\lambda_n (\ddot{q}_n + \ddot{q} - \kappa_n^2 \ddot{q}) + 2\xi_n \omega_n \dot{q}_n + \omega_n^2 q_n] \quad (10)$$

where $\alpha_2 = \int_0^L \phi(x)^2 dx$, $\alpha_4 = \int_0^L \phi(x)^4 dx$, and $\alpha_3 = \int_0^L \phi(x)^3 dx$. Since model parameters in Eq. (5) or Eq. (9) are obtained from wind tunnel test results, the dimensionless expression which is popular in wind engineering is adopted, as shown in Fig. 2, then Eqs. (7) to (10) are rewritten as

$$\eta'' + 2\xi_0 K_0 \eta' + K_0^2 \eta = m_r (\alpha_2 Y_1 \eta' - \alpha_4 Y_1 \varepsilon_N \eta'^3 + \alpha_2 Y_2 \eta) + \sum_{n=1}^N \mu_n [\lambda_n (\eta_n'' + \eta'' - \kappa_n^2 \eta'') + 2\xi_n K_n \eta_n' + K_n^2 \eta_n] \quad (11)$$

$$(\eta_n'' + \eta'') + \lambda_n (\eta_n'' + \eta'' - \kappa_n^2 \eta'') + 2\xi_n K_n \eta_n' + K_n^2 \eta_n = 0 \quad (12)$$

where $\eta(s) = q(t)/D$, $\eta_n(s) = q_n(t)/D$, $s = tU/D$; $K_0 = \omega_0 D/U$ and $K_n = \omega_n D/U$ are the reduced natural frequencies of the deck and n th TMDI; $\omega_0 = \sqrt{K/M}$ and $\xi_0 = C/2M\omega_0$ are the natural frequency and damping ratio of the deck; $m_r = \rho D^2/M$, and $\mu_n = \phi_n^2 m_n/M$. This section presents the problem formulation of the MTMDI control of the VIV in the deck.

3. The analytical solution of the deck-MTMDI system

The analytical solutions of the deck-TMD, deck-MTMD, deck-DTMD, deck-TMDI and other systems subjected the VIV, which is described by the Scanlan model, have been obtained [xx]. However, the analytical solution of the deck-MTMD system subjected to a universal VIV model is not yet available. An approximate analytical solution can help establish an objective function of the MTMDI control. In this section, the classical Krylov-Bogoliubov method [53] is adopted to solve the weak nonlinear system expressed as Eqs. (11) and (12).

According to Eq. (12), the relationship between η_n and η in the reduced frequency domain is given by

$$\eta_n = p_n + q_n i \quad (13)$$

$$p_n = (1 + \lambda_n - \kappa_n^2 \lambda_n) \frac{(K_n^2 K^2 - K^4 - \lambda_n K^4)}{(K_n^2 - K^2 - \lambda_n K^2)^2 + (2\xi_n K_n K)^2} \quad (14)$$

$$q_n = (1 + \lambda_n - \kappa_n^2 \lambda_n) \frac{-2\xi_n K_n K^3}{(K_n^2 - K^2 - \lambda_n K^2)^2 + (2\xi_n K_n K)^2} \quad (15)$$

Substituting Eq. (13) into the last item of the right side of Eq. (11) yields

$$\sum_{n=1}^N \mu_n [\lambda_n (\eta_n'' + \eta'' - \kappa_n^2 \eta'') + 2\xi_n K_n \eta_n' + K_n^2 \eta_n] = \mu_n [q_n K \eta' + (K^2 + p_n K^2) \eta] \quad (16)$$

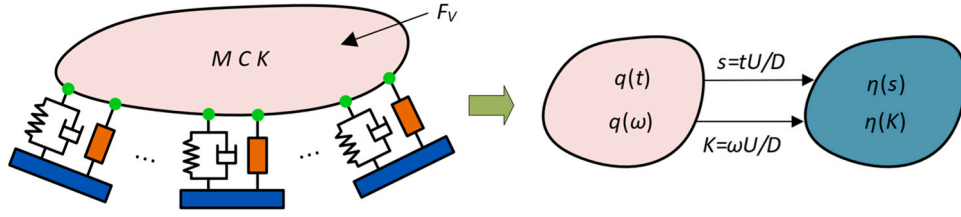


Fig. 2. The dimensionless deck-MTMDI system.

In the Krylov-Bogoliubov method, Eq. (11) is first written in a standard form

$$\eta'' + K_0^2 \eta = f(\eta, \eta') \quad (17)$$

$$f(\eta, \eta') = m_r(\alpha_2 Y_1 \eta' - \alpha_4 Y_1 \varepsilon_N \eta'^3 + \alpha_2 Y_2 \eta) - 2\xi_0 K_0 \eta' + \sum_{n=1}^N \mu_n [q_n K \eta' + (K^2 + p_n K^2) \eta] \quad (18)$$

The solutions of η and η' are assumed as

$$\eta(s) = A(s) \cos[Ks - \psi(s)] = A \cos P \quad (19)$$

$$\eta'(s) = -A(s) K \sin[Ks - \psi(s)] = -AK \sin P \quad (20)$$

where $A(s)$ and $\psi(s)$ are slowly varying functions of the defined variable s rather than constants; $P = Ks - \psi(s)$. The relationship between Eqs. (19) and (20) should satisfy the following condition

$$A' \cos P + A \psi' \sin P = 0 \quad (21)$$

According to Eq. (20), η'' can be obtained by taking the derivative of Eq. (20).

$$\eta''(s) = -A' K \sin P - AK^2 \cos P + AK \psi' \cos P \quad (22)$$

Substituting Eq. (22) into Eq. (17) yields

$$-A' K \sin P - AK^2 \cos P + AK \psi' \cos P + K_0^2 A \cos P = f(\eta, \eta') \quad (23)$$

According to Eqs. (21) and (23), two coupled first-order differential equations for $A(s)$ and $\psi(s)$ are obtained

$$A'(s) = -\frac{1}{K} [f(\eta, \eta') + A(s)(K^2 - K_0^2) \cos P] \sin P \quad (24)$$

$$\psi'(s) = \frac{1}{A(s)K} [f(\eta, \eta') + A(s)(K^2 - K_0^2) \cos P] \cos P \quad (25)$$

Since $A(s)$ and $\psi(s)$ are slowly varying functions, $A'(s)$ and $\psi'(s)$ can be considered as constants during an oscillation period $2\pi/K$. As a result, the right sides of Eqs. (24) and (25) are replaced by their means over an oscillation period,

$$A'(s) = -\frac{1}{2\pi K} \int_0^{2\pi} [f(\eta, \eta') + A(s)(K^2 - K_0^2) \cos P] \sin P dP \quad (26)$$

$$\psi'(s) = \frac{1}{2\pi K A(s)} \int_0^{2\pi} [f(\eta, \eta') + A(s)(K^2 - K_0^2) \cos P] \cos P dP \quad (27)$$

Substituting Eq. (18) into Eqs. (26) and (27), and then solving Eqs. (26) and (27), two first-order differential equations are obtained

$$A' = \frac{1}{2} m_r \alpha_2 Y_1 A - \frac{3}{8} m_r \alpha_4 Y_1 \varepsilon_N K^2 A^3 - \xi_0 K_0 A + \frac{1}{2} \sum_{n=1}^N \mu_n q_n K A \quad (28)$$

$$\psi' = \frac{1}{2K} \left[m_r \alpha_2 Y_2 + (K^2 - K_0^2) + \sum_{n=1}^N \mu_n (K^2 + p_n K^2) \right] \quad (29)$$

Furthermore, the solutions of $A(s)$ and $\psi(s)$ are obtained by solving Eqs. (28) and (29).

$$A(s) = \frac{a}{\sqrt{1 - (A_0^2 - a^2/A_0^2) e^{-(ba^2/4)s}}} \quad (30)$$

$$\psi(s) = \frac{1}{2K} \left[m_r \alpha_2 Y_2 + (K^2 - K_0^2) + \sum_{n=1}^N \mu_n (K^2 + p_n K^2) \right] s + \psi_0 \quad (31)$$

$$a = \sqrt{\frac{-8\xi_0 K_0 + 4 \sum_{n=1}^N \mu_n q_n K + 4m_r \alpha_2 Y_1}{3m_r \alpha_4 Y_1 \varepsilon_N K^2}} = \frac{2}{K\sqrt{3\alpha_4 \varepsilon_N}} \sqrt{\alpha_2 - \frac{2\xi_0 K_0 - \sum_{n=1}^N \mu_n q_n K}{m_r Y_1}} \quad (32)$$

$$b = 3m_r \alpha_4 Y_1 \varepsilon_N K^2 \quad (33)$$

where A_0 and ψ_0 are the initial displacement and phase of the deck. Finally, the approximate analytical solution of η is obtained

$$\eta(s) = \frac{a}{\sqrt{1 - (A_0^2 - a^2/A_0^2) e^{-(ba^2/4)s}}} \cos \left\{ Ks - \frac{1}{2K} \left[m_r \alpha_2 Y_2 + (K^2 - K_0^2) + \sum_{n=1}^N \mu_n (K^2 + p_n K^2) \right] s - \psi_0 \right\} \quad (34)$$

4. The optimal design of the MTMDI

In our study, the steady-state solution of η is adopted to quantify the VIV due to its simplicity. According to Eq. (34), the steady-state solution of η is obtained when $s \rightarrow \infty$

$$\eta(s \rightarrow \infty) = a = \frac{2}{K\sqrt{3\alpha_4 \varepsilon_N}} \sqrt{\alpha_2 - \frac{2\xi_0 K_0 - \sum_{n=1}^N \mu_n q_n K}{m_r Y_1}} \quad (35)$$

It is found from Eq. (35) that the contribution of the MTMDI is to increase the damping ratio of the deck, thus the steady-state vortex-induced amplitude will be reduced, and the non-steady state response will decay quickly [23–28]. The equivalent damping ratio contributed by the MTMDI is expressed as

$$\xi_{eq} = \frac{\sum_{n=1}^N \mu_n q_n K}{2K_0} = \sum_{n=1}^N \frac{\mu_n \xi_n K_n K^4 (1 + \lambda_n - \kappa_n^2 \lambda_n)}{K_0 [(K_n^2 - K^2 - \lambda_n K^2)^2 + (2\xi_n K_n K)^2]} \quad (36)$$

In the study of tuned-based dampers, the dimensionless frequency ratio is always adopted as the design parameter instead of the natural frequency. Thus, two dimensionless variables are introduced $\beta_n = K_n/K_0$ and $\gamma = K/K_0$, then Eq. (36) is rewritten as

$$\xi_{eq} = \sum_{n=1}^N \frac{\mu_n \xi_n \beta_n \gamma^4 (1 + \lambda_n - \kappa_n^2 \lambda_n)}{(\beta_n^2 - \gamma^2 - \lambda_n \gamma^2)^2 + (2\xi_n \beta_n \gamma)^2} \quad (37)$$

The dimensionless variable γ can be obtained by solving the following equation

$$\psi' = \frac{1}{2K} \left[m_r \alpha_2 Y_2 + (K^2 - K_0^2) + \sum_{n=1}^N \mu_n (K^2 + p_n K^2) \right] = 0 \quad (38)$$

Furthermore, Eq. (38) is rewritten as

$$\gamma^2 = 1 - \frac{m_r \alpha_2 Y_2}{K_0^2} - \gamma^2 \sum_{n=1}^N \mu_n \left[1 + \frac{(\beta_n^2 \gamma^2 - \gamma^4 - \lambda_n \gamma^4)(1 + \lambda_n - \kappa_n^2 \lambda_n)}{(\beta_n^2 - \gamma^2 - \lambda_n \gamma^2)^2 + (2\xi_n \beta_n \gamma)^2} \right] \quad (39)$$

The iterative technique is adopted to solve Eq. (39) to obtain multiple solutions of γ , and then the multiple ξ_{eq} are obtained by substituting multiple γ into Eq. (37) [24,44]. The smallest one among the multiple ξ_{eq} is adopted to characterize the contribution of the MTMDI. It is noted that the equivalent damping ratio expressed as Eq. (37) is established based on the steady-state oscillation assumption, thus Eq. (37) may not be suitable for characterizing the contribution of the MTMDI when ξ_{eq} is higher a critical value $\xi_{eq,c}$.

$$\xi_{eq,c} = \frac{\alpha_2 m_r Y_1}{2K_0} - \xi_0 \quad (40)$$

In the non-steady-state oscillation case, the equivalent damping ratio is identified according to the following equation by solving Eqs. (11) and (12).

$$\xi_{eq} = \delta \sqrt{\frac{1}{4n_s^2 \pi^2 + \delta^2}} - \xi_0 \quad (41)$$

where $\delta = \ln(A_0/A_n)$ is logarithmic decay rate of the peak amplitude of the deck; A_n is the peak amplitude of the deck after n_s oscillation periods. Besides, a required equivalent damping ratio $\xi_{eq,r}$ is defined to judge whether the MTMDI meets the control requirement.

$$\frac{y_{\max}}{D} = \frac{2}{K\sqrt{3\alpha_4 \epsilon_N}} \sqrt{\alpha_2 - \frac{2(\xi_0 + \xi_{eq,r})K_0}{m_r Y_1}} \quad (42)$$

where $y_{\max}$ is the maximum allowable or critical vortex-induced amplitude to keep the functionality or safety of the deck against the VIV. In the Chinese wind-resistant design specification for highway

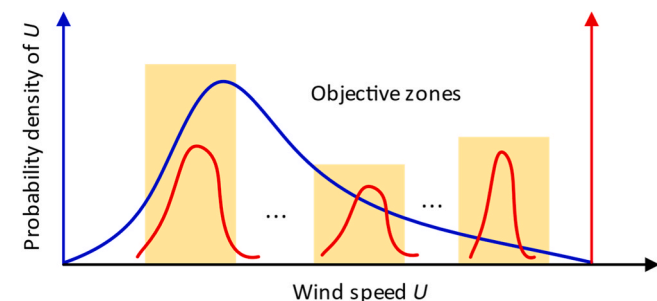


Fig. 3. The wind speed probability distribution and equivalent damping ratio.

bridges [33], $y_{\max}$ is suggested as $0.04/f$ meters, in which f is the vibration frequency. According to Eq. (42), $\xi_{eq,r}$ is expressed as

$$\xi_{eq,r} = \frac{m_r Y_1}{2K_0} \left(\alpha_2 - \frac{3\alpha_4 \epsilon_N K^2 y_{\max}^2}{4D^2} \right) - \xi_0 \quad (43)$$

Since the aerodynamic parameters Y_1 , ϵ_N , and Y_2 are functions of the wind speed, the three equivalent damping ratios ξ_{eq} , $\xi_{eq,c}$, and $\xi_{eq,r}$ are also functions of the wind speed. In other words, the MTMDI will focus on the VIV mitigation within the entire lock-in wind speed range rather than the vortex-induced resonance mitigation at a particular wind speed. Furthermore, the wind speed probability distribution is considered to fairly evaluate the risk of the VIV, as shown in Fig. 3. As a result, an objective function of the MTMDI optimization is defined as

$$F(N, \beta_n, \xi_n, \mu_n, \lambda_n, \kappa_n) = \int_{U_a}^{U_b} [\xi_{eq}(U) - \xi_{eq,r}(U)] p(U) dU \quad (44)$$

$$\xi_{eq}(U) = \begin{cases} \sum_{n=1}^N \frac{\mu_n \xi_n \beta_n \gamma^4 (1 + \lambda_n - \kappa_n^2 \lambda_n)}{(\beta_n^2 - \gamma^2 - \lambda_n \gamma^2)^2 + (2\xi_n \beta_n \gamma)^2}, & \xi_{eq} \leq \xi_{eq,c} \\ \delta \sqrt{\frac{1}{4n_s^2 \pi^2 + \delta^2}} - \xi_0, & \xi_{eq} > \xi_{eq,c} \end{cases} \quad (45)$$

where F is the objective function; $p(U)$ is the probability density function of the wind speed at the deck height; U_a and U_b are the starting and ending wind speeds, respectively.

Only limited discrete aerodynamic data are recorded in the wind tunnel tests, the aerodynamic parameters at any wind speed are obtained through the piecewise data interpolation technique, as shown in Fig. 4. In order to implement the MTMDI optimization, the Monte Carlo Simulation (MCS) is used to obtain an alternative objective function to Eq. (44).

$$F(N, \beta_n, \xi_n, \mu_n, \lambda_n, \kappa_n) = \frac{1}{Q} \sum_{q=1}^Q \xi_d(Y_{1,q}, \epsilon_{N,q}, Y_{2,q}, U_q, K_{0,q}, \xi_{0,q}) \quad (46)$$

where $\xi_d = \xi_{eq} - \xi_{eq,r}$ represents the redundancy of the equivalent damping ratio; Q is the sample size of the MCS; $K_{0,q}$, $\xi_{0,q}$, and U_q are respectively the q th samples of K_0 , ξ_0 and U according to their probability distributions; $Y_{1,q}$, $\epsilon_{N,q}$, and $Y_{2,q}$ are respectively the q th samples of $Y_1(U_q)$, $\epsilon_N(U_q)$, and $Y_2(U_q)$ according to their probability distributions; $Y_1(U_q)$, $\epsilon_N(U_q)$, and $Y_2(U_q)$ are the aerodynamic parameters at the wind speed of U_q .

If the mass ratio is small or the system uncertainty is high, the equivalent damping ratio ξ_{eq} will be not robust enough against the VIV. As a result, the objective function has to focus on reducing VIV mitigation failure rather than maximizing the redundancy of the equivalent damping ratio ξ_d . In such cases, the objective function of the MTMDI optimization is changed as

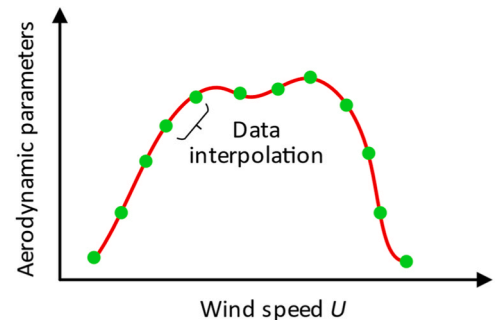


Fig. 4. The data interpolation for aerodynamic parameters.

$$F(N, \beta_n, \xi_n, \mu_n, \lambda_n, \kappa_n) = \frac{1}{Q} \sum_{q=1}^Q \frac{\text{sgn}(\xi_d) + |\text{sgn}(\xi_d)|}{2} \quad (47)$$

where $\text{sgn}(\cdot)$ is the signum function, it returns 1 and -1 for positive and negative numbers, and 0 for zero. The calculation of ξ_d is the same as that in Eq. (46).

5. Numerical example

5.1. A fictional VIV event

In this section, the MTMDI are adopted to mitigate a fictional VIV event based on wind tunnel tests of a 1/20 sectional model of the Xiangshan Harbor Bridge, with a dynamically equivalent mass of 182.178 kg. The parameters or information of the sectional model and bridge can be found in [42,43]. The natural frequencies of the sectional model and the real deck are 2.808 Hz and 0.245 Hz, respectively, accordingly the similarity ratio of the wind speed is 1.745. The aerodynamic parameter curves with the wind speed are smoothed through the piecewise cubic Hermite interpolating polynomial interpolation. Fig. 5 shows the smooth $Y_1 - U$, $\epsilon_N - U$, and $Y_2 - U$ curves. The damping ratio of the deck is assumed as 0.5%. The correlation effect of vortex-induced force along the deck span is neglected, i.e., $\alpha_2 = 1$ and $\alpha_4 = 1$. Fig. 6 shows the dimensionless vortex-induced amplitude of the deck and its maximum allowable value with the wind speed. The vortex-induced amplitude and its maximum allowable value are calculated according to Eqs. (35) and (42), respectively. It can be seen that the fictional VIV within the entire lock-in wind speed range is strong, which threatens the functionality or the VIV safety of the deck. As a result, the MTMDI are numerically adopted here to mitigate the fictional VIV in the deck.

5.2. The design parameters of the MTMDI

The total mass ratio of the MTMDI is designed as 0.8% by considering the force overload and control effect, simultaneously. The inerter cannot span a long length by considering the inerter span capacity in the deck, at the same time, the mode shape change of flexible bridges across a short span length is small, thus the inerter-span parameter κ_n varies only from 1 to 0.94. The mass, stiffness, and damping elements are installed in the mid-span of the deck. The practical implementation of the inerter for $\kappa_n < 1$ cases is realized with the help of additional outriggers [38]. According to our previous study, the control effect of the TMDI with a large inertance and a short inerter span length is not good, thus the inertance parameter λ_n is designed as a small value $\lambda_n = 3$ to reduce the static stretching from 4.14 m to 1.035 m.

In order to reduce the computational cost, the frequency ratio of each TMDI varies linearly according to $\beta_n = \beta_1 + (n-1)\beta_\Delta$, and the damping ratio of each TMDI is considered as constant. The design parameters of the MTMDI are reduced as β_1 , β_Δ , and ξ_n . The fluctuation of VIV characteristics obeys the result shown in Fig. 5. Three optimization cases are considered: **A** The wind speed probability distribution is considered

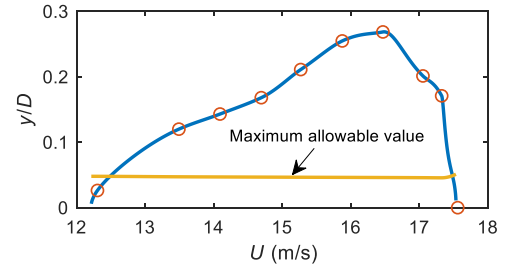


Fig. 6. The dimensionless vortex-induced amplitude of the deck and its maximum allowable value with the wind speed.

and obeys the extreme value type I distribution [45–47], in which the scale parameter and location parameter are 0.184 and 9.361 according to the wind speed data in the Xiangshan Harbor Bridge [54]; **B** The uncertainty in VIV characteristics is added into the optimization A, in which Y_1 , ϵ_N , and Y_2 are assumed to obey the normal distribution with coefficients of variation of 5%; **C** The uncertainty in deck dynamics is further added into the optimization B, in which ω_0 and ξ_0 obey the log-normal distribution with coefficients of variation of 2% and 5%. The Sobol sensitivity analysis can quantify how different variable impact the result of interest [55], but this study focuses on the control results of different MTMDIs. Therefore, the MCS is adopted here to obtain the design parameters and control results of different MTMDIs. The sample size is 10000. Table 2 presents the design parameters of different MTMDIs.

5.3. The control comparison of the case A

Fig. 7 shows the histogram of the wind speed for the Xiangshan Harbor Bridge. It can be seen that entire lock-in wind speed range between 12 m/s and 18 m/s falls in the high probability zone, which indicates that the VIV concern should be seriously considered. Fig. 8 shows the three kinds of equivalent damping ratios varying with the wind speed when $\kappa_n = 1$ in the case A. It can be seen that the large ξ_{eq} is distributed between 13 m/s and 15 m/s with high probability rather than between 15 m/s and 17 m/s with high vortex-induced amplitude, which indicates that the wind speed probability distribution seriously affects the control result. A sudden change of ξ_{eq} appears between 17.3 m/s and 17.5 m/s, because the sudden change of Y_2 cause a sharply changing vibration frequency, accordingly the TMDI or MTMDI experiences a state transformation of bad-tuning, good-tuning, and bad-tuning. The fluctuation of $\xi_{eq,r}$ with the wind speed is the same as that of the aerodynamic parameter Y_1 . The redundancy of the equivalent damping ratio ξ_d within the entire lock-in wind speed range is basically positive, which indicates the optimized TMDI and MTMDI are well mitigate the VIV. Furthermore, the MTMDI-4 has smaller ξ_d than the TMDI except for a small wind speed range between 14.3 m/s and 14.6 m/s, which indicates that an optimized MTMDI can perform better than the TMDI within the entire lock-in wind speed range. Fig. 9 shows the objective function value of the MTMDI and TMDI in the case A. The

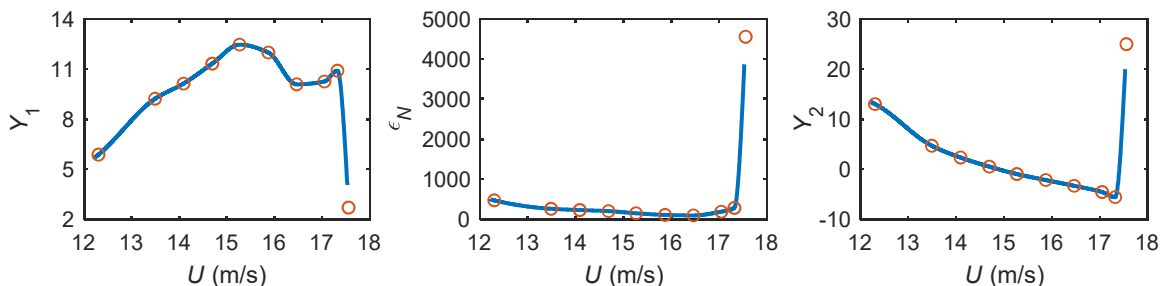
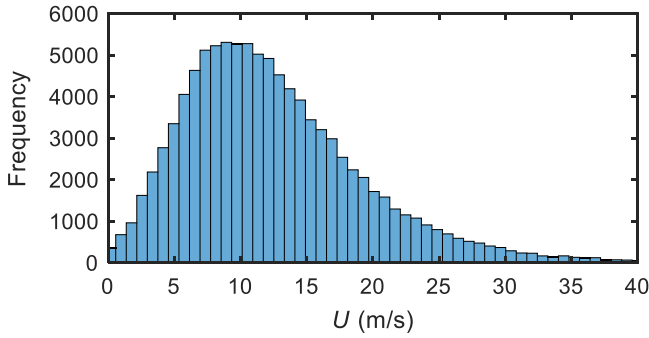


Fig. 5. The smooth $Y_1 - U$, $\epsilon_N - U$, and $Y_2 - U$ curves.

Table 2

The design parameters of the different MTMDIs.

$\sum \mu_n = 0.8\%$		$N = 1$ TMDI			$N = 2$ MTMDI-2			$N = 4$ MTMDI-4		
Case	κ_n	β_1	β_{Δ}	ξ_n	β_1	β_{Δ}	ξ_n	β_1	β_{Δ}	ξ_n
A	0.94	1.984	-	6%	1.959	0.05	4.6%	1.948	0.025	4%
	0.96	1.984	-	5.7%	1.959	0.05	4.1%	1.96	0.02	4.3%
	0.98	1.984	-	5.4%	1.96	0.05	4.2%	1.956	0.02	4.2%
	1	1.985	-	5%	1.96	0.04	4.7%	1.954	0.02	4.2%
B	0.94	1.982	-	6.3%	1.96	0.05	4.5%	1.947	0.025	4%
	0.96	1.983	-	5.7%	1.96	0.05	4.2%	1.945	0.025	3.9%
	0.98	1.986	-	5.3%	1.96	0.05	3.8%	1.95	0.025	3.5%
	1	1.984	-	5.4%	1.963	0.05	3.6%	1.952	0.025	3.7%
C	0.94	1.985	-	6.1%	1.966	0.04	5.2%	1.936	0.03	3.8%
	0.96	1.982	-	5.9%	1.968	0.05	4.2%	1.936	0.03	3.6%
	0.98	1.98	-	5.9%	1.966	0.04	4.9%	1.938	0.025	4.2%
	1	1.986	-	5.4%	1.962	0.04	4.8%	1.945	0.025	3.6%

**Fig. 7.** The histogram of the wind speed for the Xiangshan Harbor Bridge.

objective function F is calculated according to Eq. (46). It can be seen that the MTMDI-2 and MTMDI-4 both have higher objective function value than the TMDI within all the κ_n , with improvements of 8.2%~15% and 10.7%~15.6%, respectively.

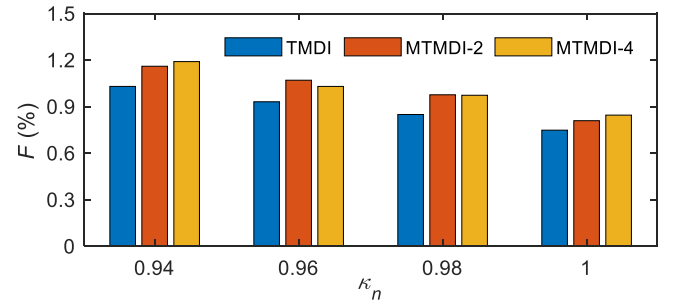
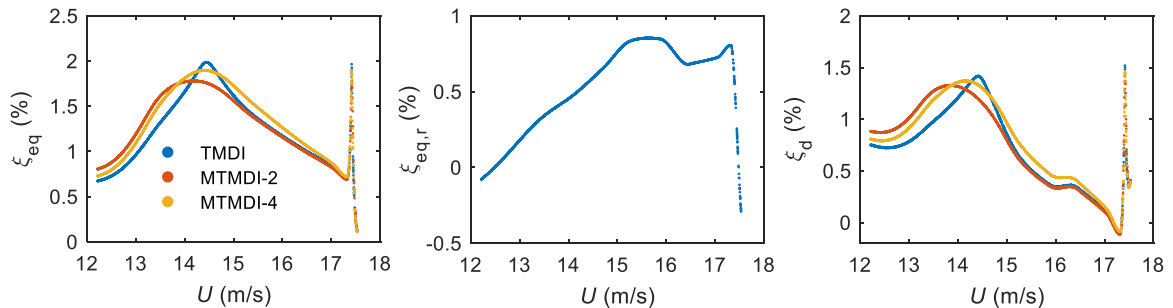
5.4. The control comparison of the case B

Fig. 10 shows the uncertain aerodynamic parameters with the wind speed. These sampling point distribution characterizes the uncertainty in VIV characteristics. Fig. 11 shows the three kinds of equivalent damping ratios varying with the wind speed when $\kappa_n = 1$ in the case B. According to Eqs. (37) and (43), the influence of aerodynamic parameters on ξ_{eq} and $\xi_{eq,r}$ is not highly nonlinear, thus the fluctuations of ξ_{eq} , $\xi_{eq,r}$, and ξ_d with the wind speed are basically the same as those shown in Fig. 8. The difference is that the superiority of the MTMDI-2 and MTMDI-4 over the TMDI between 15 m/s and 17 m/s is more significant than that shown in Fig. 8, especially for the MTMDI-4, which indicates that the MTMDI offers a better trade-off between the high probability and high vortex-induced amplitude in the VIV mitigation than the TMDI. Fig. 12 shows the objective function value of the MTMDI and TMDI in

the case B. Although the uncertainty in VIV characteristics is introduced, the reduction of the objective function value is not significant, which indicates that the influence of the uncertainty in VIV characteristics on the MTMDI control is small. Besides, the MTMDI-2 and MTMDI-4 have higher objective function value than the TMDI within all the κ_n , with improvements of 12.7%~17.8% and 15.7%~20%, respectively, which indicates that the MTMDI control is more robust than the TMDI control against the VIV uncertainty.

5.5. The control comparison of the case C

Fig. 13 shows the uncertain natural frequency and damping ratio of the deck. These sampling point distribution characterizes the uncertainty in deck dynamics. Fig. 14 shows three kinds of equivalent damping ratios varying with the wind speed when $\kappa_n = 1$ in the case C. Since the TMDI or MTMDI with small inerter span length is a frequency-sensitive damper, the fluctuations of ξ_{eq} and ξ_d with the wind speed are different from those shown in Figs. 8 and 11 after introducing the uncertainty in deck dynamics. It can be seen that ξ_{eq} is randomly distributed between 0% and 2% at any wind speed, which indicates that the uncertainty in the control result is significant. In order to clearly show

**Fig. 9.** The objective function value of the MTMDI and TMDI in the case A.**Fig. 8.** The three kinds of equivalent damping ratios with the wind speed when $\kappa_n = 1$ in the case A.

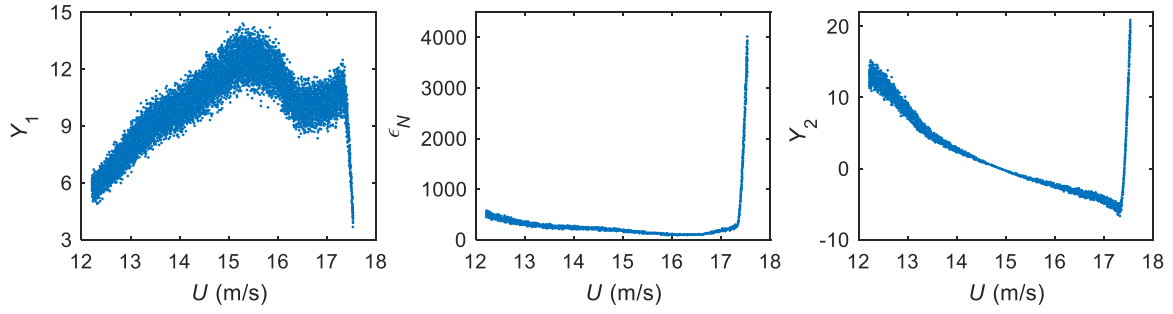


Fig. 10. The uncertain aerodynamic parameters with the wind speed.

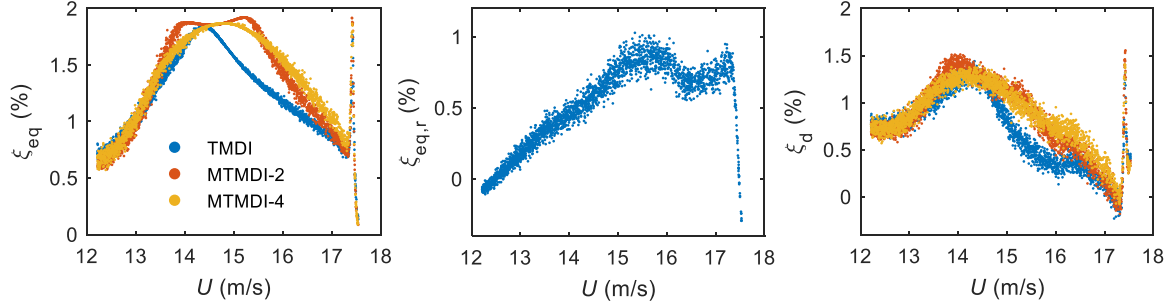


Fig. 11. The three kinds of equivalent damping ratios with the wind speed when $\kappa_n = 1$ in the case B.

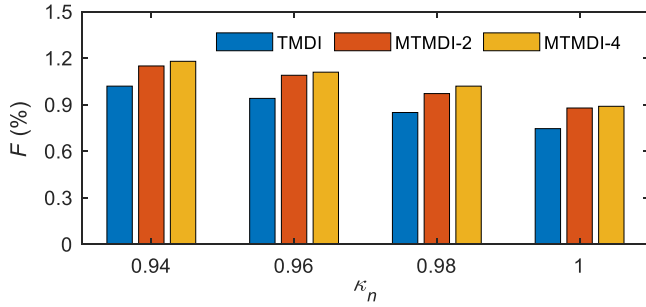


Fig. 12. The objective function value of the MTMDI and TMDI in the case B.

the comparison, the probability distribution of ξ_d when $\kappa_n = 1$ is plotted in Fig. 15. It seems that the MTMDI-4 has the best control result, followed by the MTMDI-2 and TMDI. However, the MTMDI-2 and MTMDI-4 controls are still not robust enough against the uncertainties in the VIV and deck dynamics, because many sampling points of ξ_d are negative at the wind speeds with high vortex-induced amplitude. Fig. 16 shows the objective function value of the MTMDI and TMDI in the case C. It can be seen that the objective function value decreases a lot after introducing

the uncertainty in deck dynamics, which indicates that the uncertainty in deck dynamics greatly weaken the control effect. Besides, the MTMDI-2 and MTMDI-4 have higher objective function value than the TMDI within all the κ_n , with improvements of 10%~12% and 16%~23.7%, respectively, which indicates that the MTMDI control is more robust than the TMDI control against the uncertainty in deck dynamics.

According to the results shown in Figs. 14 and 15, it can be concluded that Eq. (46) no longer provides a robust MTMDI, because the MTMDI optimization according to Eq. (46) focuses on maximizing ξ_d and neglects the VIV mitigation failure, accordingly the MTMDI easily fails in mitigating the VIV in the high system uncertainty case. Instead, Eq. (47) is used to optimize the MTMDI parameter. Table 3 presents the design parameters of the MTMDI according to Eq. (47) when $\kappa_n = 1$. It is found that the frequency ratios are more dispersed than those presented in Table 2, which indicates that the MTMDI may focus on the VIV mitigation within wider wind speed range. Fig. 17 shows the two kinds of equivalent damping ratios with the wind speed when $\kappa_n = 1$. Obviously, the distributions shown in Fig. 17 are different from those shown in Fig. 14 that, most of ξ_{eq} are distributed between 0.5% and 1.3%, and most of ξ_d are positive. Furthermore, the MTMDI shows the superiority over the TMDI in the reducing the VIV mitigation failure within the entire lock-in wind speed range. Fig. 18 shows the probability distribution of ξ_d according to Eq. (47) when $\kappa_n = 1$. It is clear that the

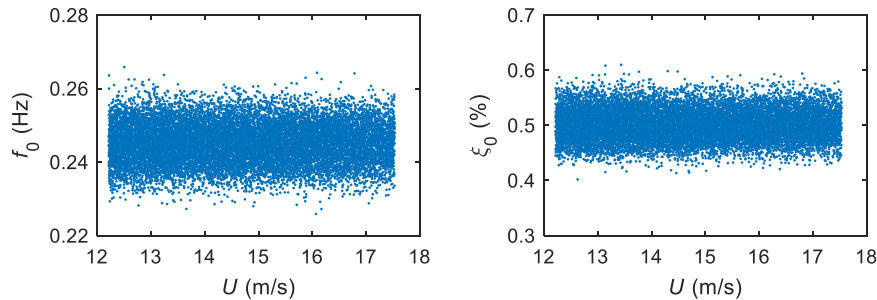


Fig. 13. The uncertain natural frequency and damping ratio of the deck.

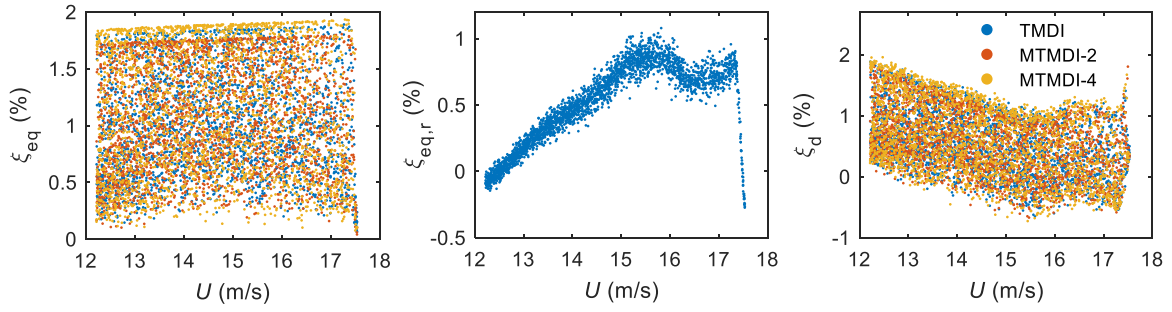


Fig. 14. The three kinds of equivalent damping ratios with the wind speed when $\kappa_n = 1$ in the case C.

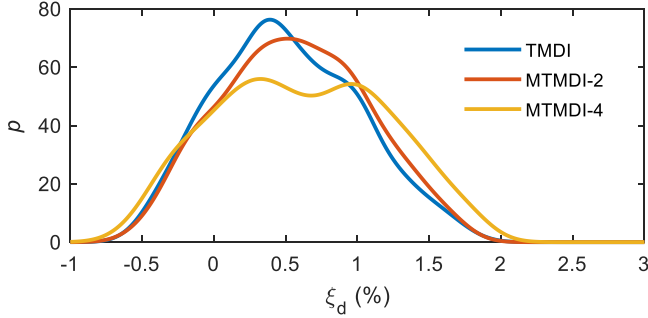


Fig. 15. The probability distribution of ξ_d when $\kappa_n = 1$.

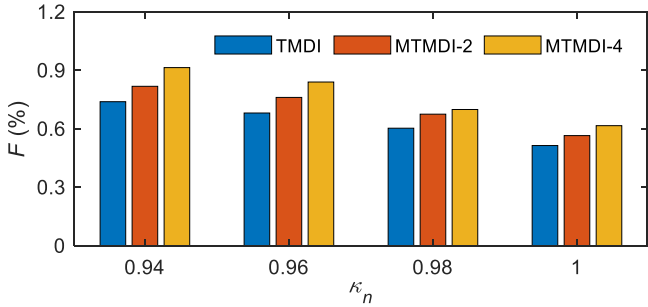


Fig. 16. The objective function value of the MTMDI and TMDI in the case C.

Table 3

The design parameters of the MTMDI according to Eq. (47) when $\kappa_n = 1$.

N	1	2	4
β_1	1.997	1.927	1.924
β_{Δ}	-	0.11	0.055
ξ_n	7.4%	5.3%	3.5%

MTMDI-4 achieves the least VIV mitigation failure, followed by the MTMDI-2 and TMDI. Compared with the result shown in Fig. 15, it is concluded that Eq. (47) is more suitable as the objective function of the

MTMDI optimization with high system uncertainty than Eq. (46).

6. Conclusions

The MTMDI was used to mitigate the VIV in flexible bridges by providing a robust equivalent damping ratio within the entire lock-in wind speed range. In the MTMDI, the inerter with equivalent high-static and low-dynamic stiffness can greatly reduce the static stretching of the auxiliary mass, to help the tuning of the very-low-frequency TMDI in a low-height installation space. The multiple damper scheme with non-identical design parameters can improve the control robustness of the TMDI with small inerter span length against the system uncertainty. A framework for optimal design of the MTMDI was proposed based on a universal nonlinear VIV mathematical model, in which the probability of occurrence of the VIV, the uncertainty in VIV characteristics, the uncertainty in deck dynamics were considered. The conclusions that can be drawn based on the numerical results are as follows,

- The relations between the aerodynamic parameters and control effect of the MTMDI are found. The fluctuation of the aerodynamic parameter Y_2 with the wind speed causes the change of vibration frequency of the deck-MTMDI system, then the equivalent damping ratio ξ_{eq} contributed by the MTMDI varies with the wind speed. The desired equivalent damping ratio $\xi_{eq,r}$ that satisfies the functionality or safety of the deck also varies with the wind speed, because the aerodynamic parameters Y_1 and ε_N those characterize the VIV intensity fluctuate with the wind speed.
- The wind speed probability distribution plays a role of weight coefficient in the MTMDI optimization by providing a fairly evaluate the risk of the VIV. The uncertainty in VIV characteristics or aerodynamic parameters does not result in a high uncertainty in the redundancy of equivalent damping ratio ξ_d . However, the uncertainty in deck dynamics causes an uncertain and poor control result due to the high frequency sensitivity induced by the small inerter span length. The defined two objective functions respectively focus on maximizing equivalent damping ratio and reducing VIV mitigation failure. The choice of the objective function depends on the mass ratio of the MTMDI and system uncertainty.

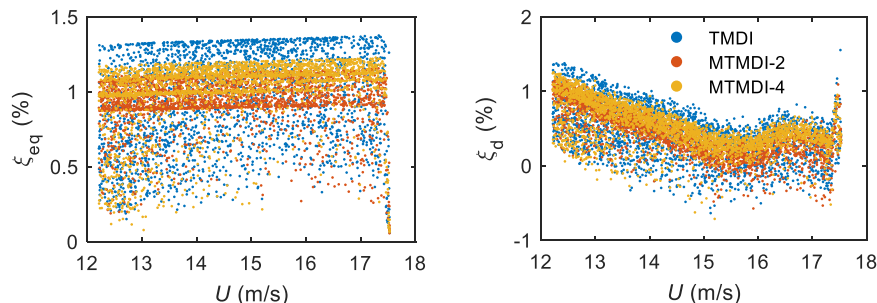


Fig. 17. The two kinds of equivalent damping ratios with the wind speed when $\kappa_n = 1$.

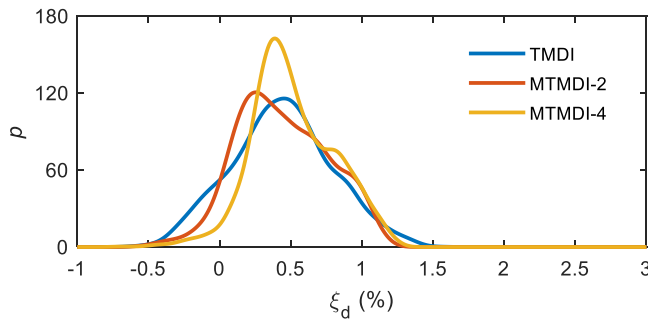


Fig. 18. The probability distribution of ξ_d according to Eq. (47) when $\kappa_n = 1$.

- The numerical results of the MTMDI control of the VIV in a flexible bridge showed that the MTMDI achieved a more robust control result than the TMDI within the entire lock-in wind speed range. Furthermore, the proposed framework for optimal design of the MTMDI can be extended to the multi-mode VIV mitigation of flexible bridges.

CRediT authorship contribution statement

Zhao-Dong Xu: Supervision, Methodology, Conceptualization. **Xiao Yan:** Software, Investigation. **Wen-Peng Xu:** Writing – original draft, Software. **Pan-Pan Gai:** Writing – original draft, Validation, Software, Methodology. **Jun Dai:** Writing – original draft, Validation, Supervision, Software, Methodology, Funding acquisition, Conceptualization. **Cen Yang:** Writing – original draft, Software, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgements

This study was financially supported by National Natural Science Foundation of China (52108442), China Postdoctoral Science Foundation (2023M741442, 2021M690619), Jiangsu Planned Projects for Postdoctoral Research Funds (2021K099A), and Major Project of Fundamental Research on Frontier Leading Technology of Jiangsu Province (BK20222006). These supports are gratefully acknowledged.

References

- [1] Sarpkaya T. A critical review of the intrinsic nature of vortex-induced vibrations. *J Fluids Struct* 2004;19(4):389–447. <https://doi.org/10.1016/j.jfluidstructs.2004.02.005>.
- [2] Williamson CHK, Govardhan R. A brief review of recent results in vortex-induced vibrations. *J Wind Eng Ind Aerodyn* 2008;96(6–7):713–35. <https://doi.org/10.1016/j.jweia.2007.06.019>.
- [3] Ge Y, Zhao L, Cao J. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. *J Wind Eng Ind Aerodyn* 2022;220:104866. <https://doi.org/10.1016/j.jweia.2021.104866>.
- [4] Owen JS, Vann AM, Davies JP, Blakeborough A. The prototype testing of Kessock Bridge: response to vortex shedding. *J Wind Eng Ind Aerodyn* 1996;60:91–108. [https://doi.org/10.1016/0167-6105\(96\)00026-8](https://doi.org/10.1016/0167-6105(96)00026-8).
- [5] Macdonald JH, Irwin PA, Fletcher MS. Vortex-induced vibrations of the second severn crossing cable-stayed bridge-full-scale and wind tunnel measurements. *Proc Inst Civ Eng -Struct Build* 2002;152(2):123–34. <https://doi.org/10.1680/stbu.2002.152.2.123>.
- [6] Burden AR. Japanese cable-stayed bridge design. *Proc Inst Civ Eng* 1991;90(5):1021–51. <https://doi.org/10.1680/icep.1991.16189>.
- [7] Seo JW, Kim HK, Park J, Kim KT, Kim GN. Interference effect on vortex-induced vibration in a parallel twin cable-stayed bridge. *J Wind Eng Ind Aerodyn* 2013;116:7–20. <https://doi.org/10.1016/j.jweia.2013.01.014>.
- [8] Smith LJ. Wind induced dynamic response of the Wye bridge. *Eng Struct* 1980;2(4):202–8. [https://doi.org/10.1016/0141-0296\(80\)90001-2](https://doi.org/10.1016/0141-0296(80)90001-2).
- [9] Ozkan E, Main J, Jones NP. Full-scale measurements on the Fred Hartman Bridge. *Proc 5th Asia-Pac Conf Wind Eng* 2001. (<http://www.jmain.sent.com.user.fm/pubs/Ozkan-Main-Jones-APCWE-V.pdf>).
- [10] Li H, Laima S, Zhang Q, Li N, Liu Z. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. *J Wind Eng Ind Aerodyn* 2014;124:54–67. <https://doi.org/10.1016/j.jweia.2013.11.006>.
- [11] Larsen A, Esdahl S, Andersen JE, Vejrum T. Storebælt suspension bridge–vortex shedding excitation and mitigation by guide vanes. *J Wind Eng Ind Aerodyn* 2000;88(2–3):283–96. [https://doi.org/10.1016/S0167-6105\(00\)00054-4](https://doi.org/10.1016/S0167-6105(00)00054-4).
- [12] Matsumoto M, Shirato H, Yagi T, Shijo R, Eguchi A, Tamaki H. Effects of aerodynamic interferences between heaving and torsional vibration of bridge decks: the case of Tacoma Narrows Bridge. *J Wind Eng Ind Aerodyn* 2003;91(12–15):1547–57. <https://doi.org/10.1016/j.jweia.2003.09.010>.
- [13] Kumarasena T, Scanlan RH, Ehsan F. Wind-induced motions of deer isle bridge. *J Struct Eng* 1991;117(11):3356–74. [https://doi.org/10.1061/\(ASCE\)0733-9445\(1991\)117:11\(3356\)](https://doi.org/10.1061/(ASCE)0733-9445(1991)117:11(3356)).
- [14] Battista RC, Pfeil MS. Reduction of vortex-induced oscillations of Rio–Niterói bridge by dynamic control devices. *J Wind Eng Ind Aerodyn* 2000;84(3):273–88. [https://doi.org/10.1016/S0167-6105\(99\)00108-7](https://doi.org/10.1016/S0167-6105(99)00108-7).
- [15] Fujino Y, Yoshida Y. Wind-induced vibration and control of Trans-Tokyo Bay crossing bridge. *J Struct Eng* 2002;128(8):1012–25. [https://doi.org/10.1061/\(ASCE\)0733-9445\(2002\)128:8\(1012\)](https://doi.org/10.1061/(ASCE)0733-9445(2002)128:8(1012)).
- [16] Corriols A.S., and Morgenthal, G. 2012. “Computational analysis of VIV observed on existing bridges.” The Seventh International Colloquium on Bluff Body Aerodynamics and Applications. Shanghai: 1356–1549. (<http://www.iawe.org/Proceedings/BBA7/A.Corriols.pdf>).
- [17] Schewe G, Larsen A. Reynolds number effects in the flow around a bluff bridge deck cross section. *J Wind Eng Ind Aerodyn* 1998;74:829–38. [https://doi.org/10.1016/S0167-6105\(98\)00075-0](https://doi.org/10.1016/S0167-6105(98)00075-0).
- [18] Wu T, Kareem A. Vortex-induced vibration of bridge decks: Volterra series-based model. *J Eng Mech* 2013;139(12):1831–43. [https://doi.org/10.1061/\(ASCE\)EM.1943-7889.0000628](https://doi.org/10.1061/(ASCE)EM.1943-7889.0000628).
- [19] Gao D, Deng Z, Yang W, Chen W. Review of the excitation mechanism and aerodynamic flow control of vortex-induced vibration of the main deck for long-span bridges: A vortex-dynamics approach. *J Fluids Struct* 2021;105:103348. <https://doi.org/10.1016/j.jfluidstructs.2021.103348>.
- [20] Fujino Y, Siringoringo D. Vibration mechanisms and controls of long-span bridges: a review. *Struct Eng Int* 2013;23(3):248–68. <https://doi.org/10.2749/101686613x13439149156886>.
- [21] Xu S, Fang G, Øiseth O, Zhao L, Ge Y. Experimental study on distributed aerodynamic forces of parallel box decks with various slot width ratios and aerodynamic countermeasures during vortex-induced vibration. *J Wind Eng Ind Aerodyn* 2023;240:105493. <https://doi.org/10.1016/j.jweia.2023.105493>.
- [22] Sarwar MW, Ishihara T. Numerical study on suppression of vortex-induced vibrations of box deck bridge section by aerodynamic countermeasures. *J Wind Eng Ind Aerodyn* 2010;98(12):701–11. <https://doi.org/10.1016/j.jweia.2010.06.001>.
- [23] Dai J, Xu ZD, Yin XJ, Gai PP, Luo Y. Parameters design of TMD mitigating vortex-induced vibration of the Hong Kong-Zhuhai-Macao Bridge deep-water non-navigable bridge. *J Bridge Eng* 2019;24(8):06019005. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001450](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001450).
- [24] Dai J, Xu ZD, Xu WP, Yan X, Chen ZQ. Multiple tuned mass damper control of vortex-induced vibration in bridges based on failure probability. *Structures* 2022;43:1807–18. <https://doi.org/10.1016/j.istruc.2022.07.028>.
- [25] Zhang M, Wu T, Øiseth O. Vortex-induced vibration control of a flexible circular cylinder using a nonlinear energy sink. *J Wind Eng Ind Aerodyn* 2022;229:105163. <https://doi.org/10.1016/j.jweia.2022.105163>.
- [26] Dai J, Xu ZD, Gai PP, Xu YW. Mitigation of vortex-induced vibration in bridges using semiactive tuned mass dampers. *J Bridge Eng* 2021;26(6):05021003. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001719](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001719).
- [27] Weber F, Mašlanka M. Frequency and damping adaptation of a TMD with controlled MR damper. *Smart Mater Struct* 2012;21(5):055011. <https://doi.org/10.1088/0964-1726/21/5/055011>.
- [28] Dai J, Xu ZD, Dyke SJ. Robust control of vortex-induced vibration in flexible bridges using an active tuned mass damper. *Struct Control Health Monit* 2022;29(8):e2980. <https://doi.org/10.1002/stc.2980>.
- [29] Wardlaw RL. The improvement of aerodynamic performance. *Aerodyn Large Bridges* 2017:59–70. <https://doi.org/10.1201/9781315136950>.
- [30] Weber F, Mašlanka M. Precise stiffness and damping emulation with MR dampers and its application to semi-active tuned mass dampers of Wolvograd Bridge. *Smart Mater Struct* 2013;23(1):015019. <https://doi.org/10.1088/0964-1726/23/1/015019>.
- [31] Sun Z, Zou Z, Ying X, Li X. Tuned mass dampers for wind-induced vibration control of Chongqi Bridge. *J Bridge Eng* 2020;25(1):05019014. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001510](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001510).
- [32] Chen ZQ, Huang ZW, Wang JH. Basic requirements of tuned mass damper for bridges and the eddy current TMD. *J Hunan Univ: Nat Sci Ed* 2013;40(8):6–10. <https://doi.org/10.3969/j.issn.1674-2974.2013.08.002>.
- [33] Chen ZQ, Huang ZW. Analysis of main factors influencing allowable magnitude of vertical vortex-induced vibration of long-span bridges. *China J Highw Transp* 2015;28(9):30–7. (<http://zgglxb.chd.edu.cn/EN/Y2015/V28/I9/30>).

- [34] Ge Y, Zhao L, Xu K. Review and reflection on vortex induced vibration of main decks of long-span bridges. *China J Highw Transp* 2019;32(10):1–18. <https://doi.org/10.19721/j.cnki.1001-7372.2019.10.001>.
- [35] Giaralis A, Petrini F. Wind-induced vibration mitigation in tall buildings using the tuned mass-damper-inerter. *J Struct Eng* 2017;143(9):04017127. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0001863](https://doi.org/10.1061/(ASCE)ST.1943-541X.0001863).
- [36] Dai J, Xu ZD, Gai PP. Tuned mass-damper-inerter control of wind-induced vibration of flexible structures based on inerter location. *Eng Struct* 2019;199:109585. <https://doi.org/10.1016/j.engstruct.2019.109585>.
- [37] Xu K, Bi K, Han Q, Li X, Du X. Using tuned mass damper inerter to mitigate vortex-induced vibration of long-span bridges: analytical study. *Eng Struct* 2019;182:101–11. <https://doi.org/10.1016/j.engstruct.2018.12.067>.
- [38] Yu H, Zhang M, Hu G. Effect of inerter locations on the vibration control performance of nonlinear energy sink inerter. *Eng Struct* 2022;273:115121. <https://doi.org/10.1016/j.engstruct.2022.115121>.
- [39] Dai J, Xu ZD, Gai PP, Hu ZW. Optimal design of tuned mass damper inerter with a Maxwell element for mitigating the vortex-induced vibration in bridges. *Mech Syst Signal Proc* 2021;148:107180. <https://doi.org/10.1016/j.ymssp.2020.107180>.
- [40] Dai J, Gai PP, Xu ZD, Huang XH. Inerter location effect on the generalized tuned mass damper inerter control. *Structures* 2023;58:105517. <https://doi.org/10.1016/j.istruc.2023.105517>.
- [41] Xu K, Dai Q, Bi K, Fang G, Ge Y. Closed-form design formulas of TMDI for suppressing vortex-induced vibration of bridge structures. *Struct Control Health Monit* 2022;29(10):e3016. <https://doi.org/10.1002/stc.3016>.
- [42] Zhu LD, Meng XL, Guo ZS. Nonlinear mathematical model of vortex-induced vertical force on a flat closed-box bridge deck. *J Wind Eng Ind Aerodyn* 2013;122:69–82. <https://doi.org/10.1016/j.jweia.2013.07.008>.
- [43] Zhu LD, Meng XL, Du LQ, Ding MC. A simplified nonlinear model of vertical vortex-induced force on box decks for predicting stable amplitudes of vortex-induced vibrations. *Engineering* 2017;3(6):854–62. <https://doi.org/10.1016/j.eng.2017.06.001>.
- [44] Dai J, Xu ZD, Gai PP. Parameter determination of the tuned mass damper mitigating the vortex-induced vibration in bridges. *Eng Struct* 2020;221:111084. <https://doi.org/10.1016/j.engstruct.2020.111084>.
- [45] Li L, Wu T, He X, Hao J, Wang H, Xu H. Reliability evaluation of vortex-induced vibration for a long-span arch bridge. *J Bridge Eng* 2018;23(5):05018002. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001225](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001225).
- [46] Zhou Q, Zheng S, Chen Z, Yang F, Liu J. Vortex-induced vibration reliability analysis in cable-stayed bridges: wind and aerodynamic uncertainties. *Struct Eng Int* 2021;1(1):51–9. <https://doi.org/10.1080/10168664.2019.1676183>.
- [47] Hu P, Hu G, Han Y, Zhang F, Tang Y. Vortex-induced force model and reliability analysis of a steel box deck with projecting slab in a cable-stayed bridge. *Int J Struct Stab Dyn* 2022;22(13):2250139. <https://doi.org/10.1142/S0219455422501395>.
- [48] Zuo H, Bi K, Hao H. Using multiple tuned mass dampers to control offshore wind turbine vibrations under multiple hazards. *Eng Struct* 2017;141:303–15. <https://doi.org/10.1016/j.engstruct.2017.03.006>.
- [49] Wang L, Shi W, Zhang Q, Zhou Y. Study on adaptive-passive multiple tuned mass damper with variable mass for a large-span floor structure. *Eng Struct* 2020;209:110010. <https://doi.org/10.1016/j.engstruct.2019.110010>.
- [50] De Domenico D, Qiao H, Wang Q, Zhu Z, Marano G. Optimal design and seismic performance of multi-tuned mass damper inerter (MTMDI) applied to adjacent high-rise buildings. *Struct Des Tall Spec Build* 2020;29(14):e1781. <https://doi.org/10.1002/tal.1781>.
- [51] Xu K, Dai Q, Bi K, Fang G, Zhao L. Multi-mode vortex-induced vibration control of long-span bridges by using distributed tuned mass damper inerters (DTMDIs). *J Wind Eng Ind Aerodyn* 2022;224:104970. <https://doi.org/10.1016/j.jweia.2022.104970>.
- [52] Zhang M, Xu F, Øiseth O. Aerodynamic damping models for vortex-induced vibration of a rectangular 4: 1 cylinder: Comparison of modeling schemes. *J Wind Eng Ind Aerodyn* 2020;205:104321. <https://doi.org/10.1016/j.jweia.2020.104321>.
- [53] Ehsan F, Scanlan RH. Vortex-induced vibrations of flexible bridges. *J Eng Mech* 1990;116(6):1392–411. [https://doi.org/10.1061/\(ASCE\)0733-9399\(1990\)116:6\(1392\)](https://doi.org/10.1061/(ASCE)0733-9399(1990)116:6(1392)).
- [54] Qin XZ. Master Thesis: Wind-Induced Damage Risk Analysis of Large Span Highway Cable-Stayed Bridges. Dalian China: Dalian Maritime University; 2023. (<https://doi.org/10.26989/d.cnki.gdlhu.2022.000147>).
- [55] Hu X, Fang G, Ge Y. Uncertainty propagation of flutter derivatives and structural damping in buffeting fragility analysis of long-span bridges using surrogate models. *Struct Saf* 2024;106:102410. <https://doi.org/10.1016/j.strusafe.2023.102410>.